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Photovoltaic/thermal (PV/T) solar system: Experimental measurements, performance analysis and economic assessment

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ABSTRACT

The main scope of the paper is studying the potential offered by photovoltaic/thermal (PV/T) solar system in Central Italy. The investigation is performed according to two approaches. Firstly, an extensive monitoring of a PV/T solar system was carried out in Forlì (Italy). The PV/T solar plant was remotely monitored and data on environmental conditions, module temperature, cooling fluid flowrate and temperatures and both electric and thermal power production were continuously acquired and stored. The results showed that i) PV cooling produces a negligible increasing in the electric yield (1–3%) and ii) the PV/T solar system is able to reach a simultaneously generation of electricity (835 kWh/m^2) and heat (1600 kWh/m^2) at a mean outlet temperature over 40 °C. The second approach is the estimation of monthly/annual thermal performances of the PV/T solar system under the monitored climate condition by using ISO 9806:2013 model.

The results showed that the PV/T solar system can produce about 1362 kWh/year of electricity for kWp installed, while yearly heat generation may vary between 443 and 267 kWh/m^2 depending on mean inlet temperature of the cooling fluid. The paper also reports a techno-economic comparison of the PV/T solar plant with separate photovoltaic and flat plate solar thermal.

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1. Introduction

Photovoltaic (PV) modules and solar thermal collectors are being installed in residential buildings and in some industrial utilities to respectively produce electrical and thermal energy. Both technologies have been widely applied and demonstrated to be reliable and effective. In particular, solar thermal collector is one of the most cost effective renewable energy technologies and has already demonstrated a huge market potential. The global solar thermal market has been continuously growing since the beginning of the 1990s, and in Europe the total installed capacity reached 33 GWth in 2014 [1]. Solar thermal collectors can be utilized for various purposes: domestic hot water generation, building heating, solar assisted cooling and industrial process heating. Nevertheless, PV system has recently reached the status of a technically and commercially mature technology, able to generate electricity using solar energy with competitive costs (grid parity) if compared to traditional fossil fuel technologies. The PV worldwide market has grown rapidly in the last few years [2], also thanks to specific national and transnational incentive programs.

Photovoltaic/Thermal (PV/T) technology is a hybrid technology combining PV and solar thermal components into a single module to simultaneously generate electricity and heat. The integration of PV and solar thermal technology in one device enhances solar energy conversion efficiency and also achieves optimized use of surfaces. The idea of integrating PV and solar thermal technology together is not a new one, since it was developed starting back in the '70s [3]. On the other hand, the industrial development of PV/T technology is quite recent, since it was necessary to wait until PV and solar thermal systems reached techno-economic maturity. There is a considerable amount of literature on PV/T prototypes and experimental device modelling and performance analysis [4–6].

Different kinds of PV/T prototypes have been tested in different locations and with different cooling systems [7–17], including comparisons with PV systems without active cooling, as well as considering different kinds of fluids, in particular water and/or air. Numerical methods were also developed to foresee performance and production yields of different PV/T solar systems, including prototypes (sheet and tubes [7], building-integrated [10], glazed and unglazed collector with and without selective coatings [12],

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Nomenclature

A_G	Gross surface area of PV/T modules (m^2)
A_A	Absorber area of PV/T modules (m^2)
A_P	Aperture area (m^2)
a_1	Linear coefficient of thermal dispersion (W/m^2K)
a_2	Quadratic coefficient of thermal dispersion ($W/m^2\ K^2$)
b_0	Incident angle constant (dimensionless)
C	Thermal capacity of solar collector (J/K)
$C_0^{O\&M}$	Operation and maintenance costs (€/m^2)
c_F	Heat capacity of the cooling fluid (J/kgK)
DIN	Department of Industrial Engineering
e	Inflation rate (%)
E	DC electrical energy (Wh)
E_y	Yearly electric and thermal energy production (kWh/year)
E_{th}	Thermal energy (Wh)
F_0	Initial investment (€/kWp)
G	Measured global solar radiation (W/m^2)
G_{STC}	Global solar radiation at STC (W/m^2)
HE	HEnergia
k_Θ	Incident angle modifier function (%)
i	Discount rate (%)
I	Solar radiation energy per surface unit (Wh/m^2)
IAM	Incident Angle Modifier
LCOE	Levelized Cost Of Energy (€/kWh)
m_F	Mass flowrate of the cooling fluid (kg/s)
n	Period of investment evaluation (number of years)
NOCT	Nominal Operating Cell Temperature (°C)
O&M	Operation and maintenance
P_{DC}	DC power (W)
P_h	Thermal power (W)
P_{nom}	Nominal DC power (or peak power) (W)
PR	Performance ratio (%)
PV	Photovoltaic
PV/T	Photovoltaic/Thermal
$r_{O\&M}$	Operation and maintenance escalation rate (%/year)
t	Time (year)
T_a	Ambient temperature (°C)
TC	Thermal Collector
T_{in}	Inlet temperature of the cooling fluid (°C)
T_m	Mean temperature of the cooling fluid (°C)
T_m^*	Reduced temperature (m^2 °C/W)
T_{mod}	Temperature of PV module (°C)
T_{out}	Outlet temperature of the cooling fluid (°C)
T_{out_m}	Annual average of outlet temperature of the cooling fluid (°C)
STC	Standard Test Condition
Y_A	Array yield (Wh/W)
Y_R	Reference yield (Wh/W)
η_0	Peak thermal efficiency (%)
η_h	Thermal efficiency (%)
η_{STC}	DC efficiency at STC (%)
η_{DC}	DC efficiency of PV modules (%)
γ	Temperature coefficient (%/°C)
Θ	Incident angle (°)

improved laminated absorber [15], low emissivity coatings [18], air PV/T [19]) and commercial modules (Energies-Sol [20,21], Solimpeks [22]). The effect of active cooling on different kinds of PV technologies has also been studied [16,17]. Finally, several economic assessments of PV/T technology have been carried out (plant installed in Honk Kong [10], Germany [12], Thailand [17], UK [21]), mainly regarding payback time in comparison with traditional energy sources. But apart from performance modelling and market and cost forecasts, there are few reliable data on the techno-economics comparison of commercial PV/T devices with separate PV and solar thermal collectors (plant installed in Germany [12], different locations in France [15] and USA [23], Switzerland [24]). A different approach to generate heat and electricity through one solar device benefits from Seebeck cells integration [25,26], instead of PV cells, but this technology is still far from being commercially available due to Seebeck cells application drawbacks, which are high costs and low efficiency [27]. So, market competitors of PV/T technology are separate PV and solar thermal collectors.

The main objective of this paper is to measure and assess the potential offered by the use of a commercial PV/T solar collector to supply electricity and domestic hot water simultaneously in Central Italy (or, more in general, Central-South Europe). The paper's methodology firstly consists in analysing the experimental measurements produced by the PV/T solar plant (Fig. 1) hosted in the applied research centre on renewable energy called "HEnergia" (HE), located in Forlì (Italy, 44°13′21″N-12°02′27″E). HE was inaugurated at the end of September 2013 by Hera S.p.A., which is one of the most important Italian multiutility companies. HE was designed and created in collaboration with the Department of Industrial Engineering (DIN) of the University of Bologna and hosts different kinds of renewable energy plants [28].

The PV/T solar plant performances have been monitored with and without active cooling and by relating to real environmental



Fig. 1. Picture of HEnergia research centre: the PV/T solar system is in the foreground.

conditions. The paper also ascertains whether the thermal performance of the PV/T solar plant can be accurately predicted by applying the ISO 9806:2013 standard model. On the basis of the real environmental conditions measured and by applying the same ISO standard, the paper then estimates the yearly thermal yield production of the PV/T solar plant. The originality of the paper lies in the fact that i) it covers an energy study performed to obtain thermal and electric outputs of a commercial PV/T solar plant under real environmental conditions, and ii) it includes a validation and application of the ISO 9806:2013 model to predict the annual thermal yield of the PV/T solar plant.

Finally, the paper presents a comparison between the performances of PV/T and of separate, but simultaneous, PV and solar thermal collector technologies (in particular, flat plate solar

thermal collectors) to verify improvement and to identify the drawbacks of PV/T, including relevant suggestions for designers. The comparison includes an economic assessment carried out through the Levelized Cost of Energy (LCOE) method to estimate the cost limits under which PV/T solar technology may prove as reliable and competitive as separate PV and solar thermal collectors.

2. Material and methods

2.1. The PV/T solar plant and monitoring system

In the spring of 2013, eight different PV systems and one PV/T solar system were installed at HE [28]. The PV/T solar plant consists of four Anaf Solar modules. Table 1 summarises the Anaf Solar module main characteristics, both electric and thermal. In particular, the PV/T electric yield refers to the Standard Test Condition (STC), i.e. 1000 W/m^2 of global solar radiation and 25 °C of PV module temperature, while the thermal properties are defined according to EN 12975-1:2006 test methods. The modules were installed at ground level, with a horizontal inclination angle of 30° and South-exposed.

The PV/T solar modules are electrically connected in series in one string, while they are thermally connected in parallel to avoid different working conditions for the PV modules. The Anaf Solar module has an aluminium collector on the rear. The collector transfers the heat generated by solar radiation and by the Joule effect (produced by PV cells) to a closed thermal system, which works with a mixture of water and glycol (Fig. 2).

The PV section of the plant feeds energy via standard inverter into the grid and it is monitored and logged on the DC and AC side of the inverter (Power One inverter, model Aurora 3.0). The PV monitoring activities are done in accordance with the international standard IEC 61724:1998. For every string, current and voltage (DC Wattmeter Electrex Femto) are monitored on the DC side. The power on the AC side of the inverter (AC Wattmeter Electrex Femto) is also measured. The plant is also equipped with a PT100 temperature sensor (Omega SA1-RTD-4W-120), installed on the rear of one of the PV/T solar plant modules.

The Piping and Instrumentation Diagram (P&ID) of the solar thermal section of the PV/T solar plant is shown in Fig. 3. The heat is exchanged from the PV/T solar thermal system by a primary cooling circuit. Both inlet and outlet temperatures (Siemens PT100) of the cooling fluid as well as cooling fluid volumetric flowrate (Siemens Sitrans FM MAG) are measured in the primary circuit. The cooling fluid of the primary circuit is a 30% vol. mixture of propylenic glycol and water (the main characteristics of the cooling fluid as a temperature function are shown in Fig. 4). The primary cooling circuit transfers the heat to a secondary circuit through a plate heat exchanger. The heat is then stored in an 800 L tank. There are no utilities connected to the tank, so a dry cooler can be used as heat sink, if needed.

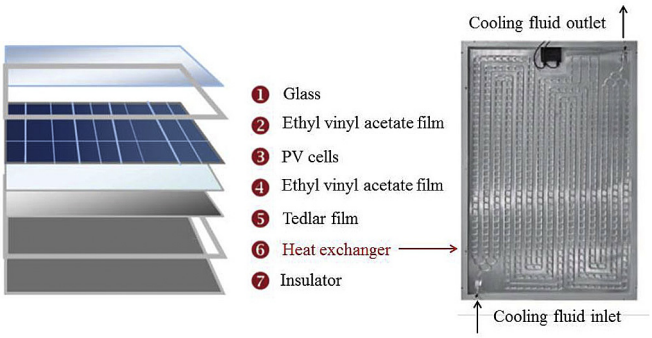


Fig. 2. Construction scheme of the Anaf Solar PV/T module.

The PV/T solar plant is automatically regulated by the following procedure: the pump of the primary circuit turns on when solar radiation is measured. Once turned on, the primary circuit pump continues working only if, after 5 min, T_{out} is 5 °C higher than T_{in}, otherwise the pump is switched off and then restarted after 15 min. The pump of the secondary circuit is turned on if the temperature in the storage tank is lower than T_{out}. Finally, the dry cooler is turned on when the temperature inside the storage tank reaches a set-up value. The primary cooling circuit reproduces a circuit for domestic hot water production: Anaf Solar module cooling circuit is thus designed not to optimize electric energy production from PV modules, but to produce domestic hot water from PV module cooling at relatively high temperatures (over 35 °C).

Furthermore, environmental conditions such as solar global radiation (Kipp & Zonen pyranometer CMP11), ambient temperature and humidity (LSI Lastem thermohygrometric sensor DMA675), wind speed and direction (LSI Lastem wind speed and direction sensor DNA721) are measured through a local meteorological station, installed on the HE roof. Each sensor mentioned above produces signals in 4–20 mA. Signals coming from the PV/T solar plant are sent to the HE datalogger, while signals coming from the local meteorological station are sent to a dedicated datalogger. Data from the dataloggers are acquired and stored every 5 s in a PC via a LAN network. Data are managed through MySQL Workbench on a PC that can be remotely accessed via web.

All the installed instruments have a maximum measuring error of 0.5%, with the exception of temperature transmitters, which are all Class A (according to DIN EN 60 751:1995): this means that when measuring a temperature of about 40 °C, a maximum measuring error of 0.5% is allowed. The architecture of the data

Table 1
PV/T Anaf Solar module main characteristics.

Module characteristic	Symbol	Value
Dimensions (mm × mm × mm)	–	1652 × 1000 × 45
Gross area (m^2)	A_G	1.652
Absorber area (m^2)	A_A	1.460
Weight (kg)	–	34
Electric power at STC (W)	P_{nom}	230
Electric efficiency at STC (%)	η_{STC}	13.9
Temperature coefficient (%/°C)	γ	−0.45
Nominal Operating Cell Temperature (°C)	NOCT	45
Peak thermal efficiency (%)	η_0	0.650
Linear coefficient of thermal dispersion (W/m^{2}K)	a_1	14.405
Quadratic coefficient of thermal dispersion (W/m^2 K^2)	a_2	0
Incident Angle Modifier at $\Theta = 50°$ (%)	$k_\Theta(50)$	96

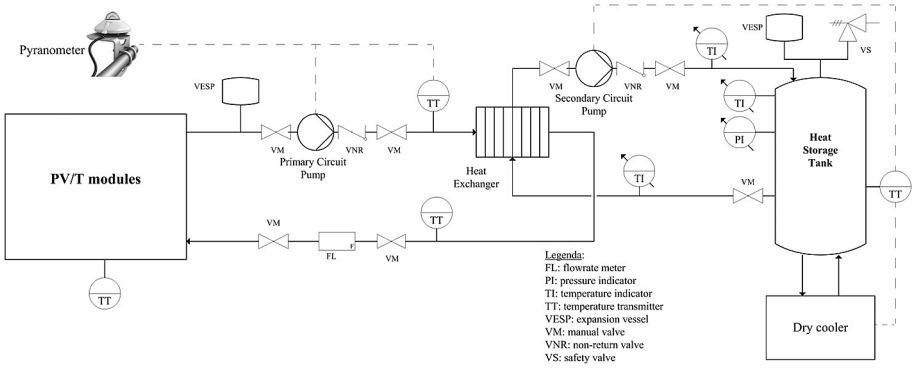


Fig. 3. Piping and Instrumentation Diagram (P&ID) of the solar thermal section of the PV/T solar plant.

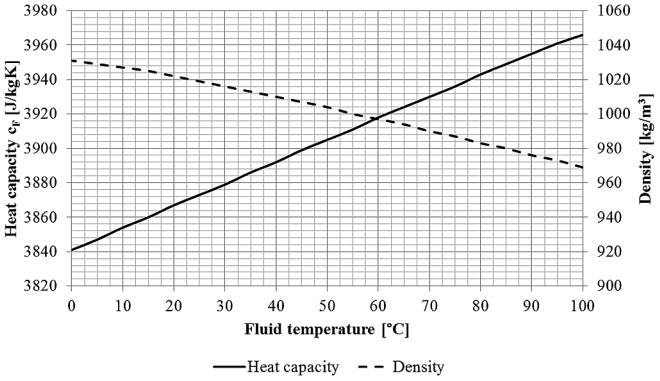


Fig. 4. Main characteristics of the cooling fluid (30% vol. mixture of propylenic glycol and water) as a temperature function.

acquisition system is shown in Fig. 5.

2.2. Electric performance indexes

The main electric index, used in the absence of direct measurements, is the efficiency at Standard Test Condition (STC) η_{STC} (Eq. (1)), defined as the ratio between the nominal DC power (or peak power) P_{nom} at STC (measured in Watts) and the product of the surface area of modules A_G (expressed in square meters) by the STC solar radiation G_{STC} (measured in W/m^2).

$$\eta_{STC} = \frac{P_{nom}}{A_G \cdot G_{STC}} \tag{1}$$

When PV modules work in a real environment, the measured DC power (expressed in Watts), P_{DC}, can be very different from the nominal one due to variations in temperature of module T_{mod} (expressed in °C) and/or radiation G (expressed in W/m^2). The DC efficiency η_{DC} of PV modules can be defined as the ratio between measured DC power P_{DC} and the product of the surface area of modules A_G with the measured solar radiation G (Eq. (2)).

$$\eta_{DC} = \frac{P_{DC}}{A_G \cdot G} \tag{2}$$

These electric indexes evaluate the module performances instantaneously. The IEC standard 61724:1998 defines performance indexes referred to a period of time: in this case, instead of electrical power and solar radiation the corresponding energy values in

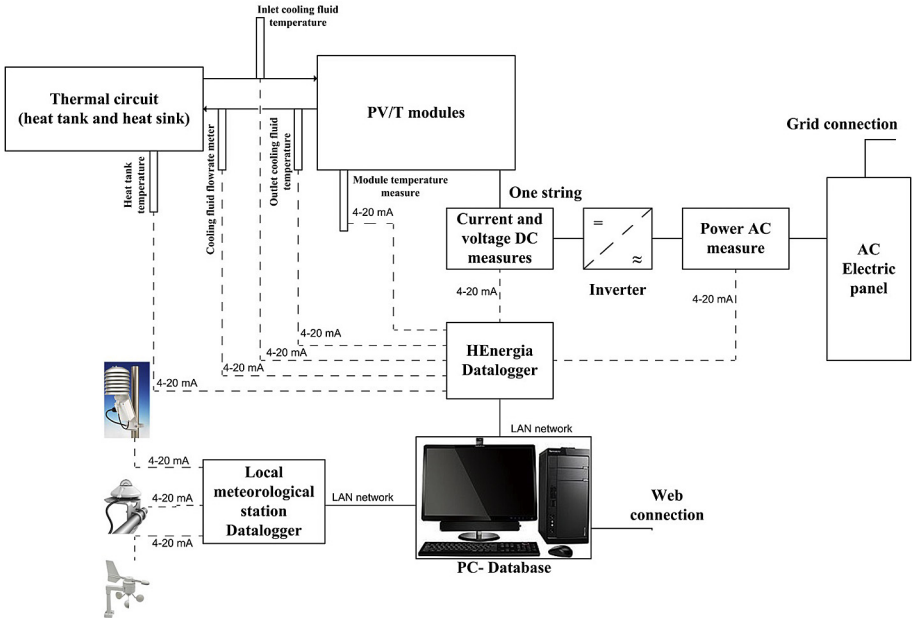


Fig. 5. Layout of the data acquisition system.

the defined period of time (day, month, year) have to be evaluated. The array yield Y_A is defined as the ratio between the DC electrical energy produced in a defined time interval E (expressed in Wh) and P_{nom}. So, Y_A is expressed in Wh/W. This electric index can also be interpreted as the number of hours in which PV modules work at their peak value in the defined time interval.

The reference yield Y_R is defined as the ratio between the solar radiation energy per surface unit I evaluated in the considered time interval (expressed in Wh/m^2) and G_{STC}. Thus, Y_R is also expressed in Wh/W. Finally, the performance ratio PR is defined as the ratio between Y_A and the reference yield Y_R (Eq. (3)).

$$PR = Y_A/Y_R, \quad \text{with} \quad Y_A = E/P_{nom} \quad \text{and} \quad Y_R = I/G_{STC} \tag{3}$$

2.3. Thermal performance indexes and ISO 9806 model

ISO 9806:2013 standard covers performance, durability and reliability testing of almost all the solar thermal collector types available on the market, including PV/T solar systems. At present, only solar thermal collector types covered by the ISO 9806:2013 standard can obtain the Solar Keymark certificate, which is a voluntary third-party certification mark for solar thermal products, demonstrating to end-users that a product conforms to the relevant European standards and fulfils additional requirements. Anaf Solar modules have no Solar Keymark certificate.

Accordingly to ISO 9806:2013, a peak condition can also be defined for solar thermal modules as for PV modules. First of all, the thermal peak condition occurs when no heat loss is present. The absence of thermal losses can be reached when the reduced temperature T_m^*, which is defined as in Eq. (4), is equal to zero. When $T_m^* = 0$, the mean temperature of the cooling fluid T_m (defined as in Eq. (5)) and the ambient temperature T_a are the same.

$$T_m^* = \frac{T_m - T_a}{G} \tag{4}$$

$$T_m = \frac{T_{out} + T_{in}}{2} \tag{5}$$

The second requirement for being in a thermal peak condition is that the direct solar radiation has to be normal to the collector surface (incident angle $\Theta = 0°$). ISO 9806:2013 defines four parameters that characterize solar thermal collector performances. The first one is the peak thermal efficiency η_0, which is the ratio between the thermal power produced by a solar thermal collector at a given solar radiation G when $T_m^* = 0$ and G is perpendicular to the gross area A_G of the collector surface. Then, two more parameters are defined to characterize the solar thermal collector tendency to dissipate heat: the linear coefficient of thermal dispersion a_1 (measured in W/m^2 K) and the quadratic coefficient of thermal dispersion a_2 (measured in W/m^2 K^2). In the case of Anaf Solar modules, these three parameters (η_0, a_1, a_2) refer to the absorber area A_A, instead of referring to the gross area A_G, as usually happens.

Finally, the fourth parameter which characterizes solar thermal collector performance is the Incident Angle Modifier (IAM) function k_Θ, which expresses the percentage variance in output performance of a solar collector as the incident angle Θ of solar radiation changes in relation to the surface of the collector. In the case of a flat plate solar thermal collector, ISO 9806:2013 gives the following equation (Eq. (6)) to determine k_Θ trend, where b_0 is a constant.

$$k_\Theta = 1 - b_0 \cdot \left(\frac{1}{\cos\Theta} - 1\right) \tag{6}$$

The thermal efficiency of the solar thermal collector η_h can thus be expressed as in Eq. (7) for steady-state conditions. A more complex equation for the η_h estimation in quasi-dynamic conditions is given by ISO 9806:2013, taking into account also the wind effect and the PV/T solar system thermal inertia. On the other hand, if the wind speed is in the range 2–4 m/s (as measured in HE), wind speed effects can be ignored. The quasi-dynamic and steady-state models are considered to give comparable results for most collector designs [29]: for example, vacuum tubes with cylindrical absorbers or semi concentrating collectors need a more complex modelling. PV/T solare thermal inertia can thus be ignored for the Anaf Solar modules. The thermal power P_h can therefore be predicted by Eq. (8).

$$\eta_h = \eta_0 \cdot k_\Theta - a_1 \cdot T_m^* - a_2 \cdot G \cdot \left(T_m^*\right)^2 \tag{7}$$

$$P_h = A_A \cdot G \cdot \eta_h = A_A \cdot G \cdot \left[\eta_0 \cdot k_\Theta - a_1 \cdot T_m^* - a_2 \cdot G \cdot \left(T_m^*\right)^2\right] \tag{8}$$

The thermal power P_h produced by the PV/T solar plant can be computed by Eq. (9), once mass flowrate m_F and heat capacity c_F of the cooling fluid are known, and the PV/T inlet T_{in} and outlet T_{out} temperatures of the cooling fluid are measured.

$$P_h = m_F \cdot c_F \cdot (T_{out} - T_{in}) = 2 \cdot m_F \cdot c_F \cdot (T_m - T_{in}) \tag{9}$$

So, by combining Eqs. (8) and (9), it is possible to state that once environmental conditions (T_a and G), solar thermal collector characteristics (η_0, k_Θ, a_1, a_2, A_A) and working conditions (m_F, c_F and T_{in}) are known, T_m (or T_{out}) can be estimated through Eq. (10), and then P_h can also be estimated. Eq. (10) can be identified as the "ISO 9806 model" and has been applied in the paper to predict thermal yield of both PV/T and flat plate solar collectors.

$$T_m = T_a + G\frac{-\left(\frac{a_1}{a_2 G} + \frac{2\dot{m}c_p}{AGa_2}\right) + \sqrt[2]{\left(\frac{a_1}{a_2 G} + \frac{2\dot{m}c_p}{AGa_2}\right)^2 - 4\left(\frac{2\dot{m}c_p T_a}{AG^2 a_2} - \frac{\eta k}{Ga_2} - \frac{2\dot{m}c_p T_{in}}{AG^2 a_2}\right)}}{2} \tag{10}$$

2.4. PV/T solar plant comparison with separate PV and solar thermal plants

The paper aims to compare the techno-economic performance of the tested PV/T solar plant with separate PV and solar thermal plants. The comparison is carried out by considering two different plant configurations: the first one is a PV/T solar plant with 3.2 kWp el (characteristics in Table 2), while the second one includes both PV panels and solar thermal collectors, opportunely sized to produce the same annual amount of electric (net energy used for cooling fluid pumping) and thermal energy (at the same annual mean temperature) as the PV/T solar plant.

Table 2
3.2 kWp el PV/T Anaf Solar plant main characteristics.

Module characteristic	Symbol	Value
Peak electric power output (kWp el)	P_{nom}	3.220
Number of modules	–	14
Gross area (m^2)	A_G	23.128
Absorber area (m^2)	A_A	20.440
Cooling fluid flowrate (kg/s)	m_F	0.224
Pressure drop (mbar)	–	160

An ideal plant to produce domestic hot water was designed to compare the PV/T solar plant performance with the performance of a solar thermal plant. Fig. 6 shows the P&ID of the ideal plant: the water coming from the water network is heated up to 40 °C by the heat exchanger in a heat storage tank. Thermal energy produced by PV/T and solar thermal modules is stored in the tank. This solution was chosen instead of the direct domestic hot water storage to avoid legionella formation problems [30]. A methane boiler is present as back-up and/or integration unit.

Two hypotheses were proposed for the ideal plant control system: the first one is that the cooling fluid inlet temperature T_{in} is supposed to be constant over time. Fluid inlet temperature variation does not affect PV/T module performances at constant framework conditions like solar radiation and ambient temperature (see Fig. 9). Moreover the hypothesis simplifies the comparison between the PV/T and the solar thermal collector without compromising results replication. The second hypothesis is that the cooling fluid pump is activated when a solar global radiation higher than 300 W/m^2 is measured: if the pump is on and the solar radiation falls below 300 W/m^2, the pump is switched off. This second hypothesis is defined to optimize the operation of the cooling fluid pump. The limit of 300 W/m^2 was chosen on the basis of PV/T solar plant experimental monitoring and data analysis. Moreover, the cooling fluid mass flowrate is supposed to be constant and equal to the maximum value suggested by PV/T solar module and solar thermal collector producers.

PV/T and PV electric performances were compared on the basis of experimental measurements. Instead, PV/T and solar thermal collector performances were computed through Eq. (10), starting from environmental data recorded by the local meteorological station.

Solar thermal collectors can be classified into five core technology types [31–33], which most collectors derive from: (i) unglazed collectors, (ii) glazed flat plate collectors, (iii) massive solar thermal collectors, (iv) evacuated tube collectors and (v) concentrated solar collectors. Flat plate collectors account for 26.4% of the global and 84.9% of the European installed solar thermal capacity [33]. So, the comparison was made with a flat plate thermal solar collector, having the characteristics of a baseline product, since the system has the sole purpose of producing domestic hot water, thus not requiring a high performance collector (i.e. evacuated tube collector). Table 3 summarises the characteristics of the solar thermal collector identified as representative of low-cost solar thermal collectors available on the European market, which is the VMF 2.0 model produced by Ferroli (Italy). The VMF 2.0 model consists of an aluminium collector case; the presence of a highly selective copper absorber and a tempered anti-reflective prismatic glass with low iron content allows a good compromise to be reached in terms of thermal efficiency and module cost. Moreover, the VMF 2.0 is certified by Solar Keymark.

The whole aperture area A_P of the solar thermal collector plant

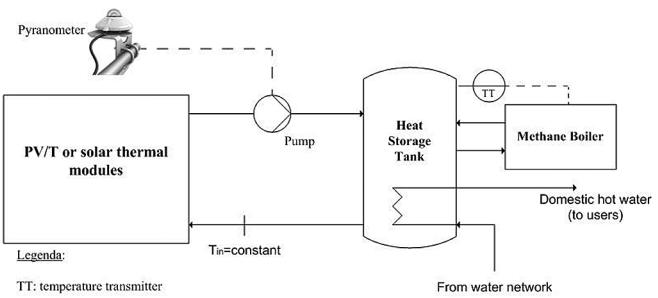


Fig. 6. P&ID of the ideal plant for domestic hot water production.

was computed through an iterative method in order to reach the same mean annual cooling fluid outlet temperature and thermal energy production as the PV/T solar plant. The electric energy consumed for cooling fluid pumping was also computed. Then, a separate PV plant was sized to reach the same electric energy production as the PV/T solar plant, net energy expenditure for cooling fluid pumping. It was planned to use the same PV technology as the PV/T solar plant.

2.5. Economics considerations

An economics analysis of the PV/T solar plant was carried out through the application of the Levelized Cost Of Energy (LCOE) method [28,34]. The LCOE methodology is an abstraction from reality and it is used as a benchmark to assess the cost-effectiveness of different energy generation technologies. The method considers the lifetime generated energy and costs to estimate a price per unit energy generated. In this case, the LCOE for the production of both electric and thermal energy through a PV/T solar plant will be compared with the LCOE for the production of the same energy (electric and thermal) through separate PV and solar thermal modules. The formula of LCOE can be expressed as in Eq. (11):

$$LCOE = \frac{F_0 + \sum_{t=1}^{n} C_0^{O\&M} \cdot (1 + e + r_{O\&M})^t \Big/ (1+i)^t}{\sum_{t=0}^{n} E_y \Big/ (1+i)^t} \quad (11)$$

where t (years) is time, n (number of years) is the time period considered for the investment evaluation (which will be assumed equal to both depreciation and technical life time of plants for treatment simplicity purpose), i (%) is the discount rate and E_y (kWh/year) is the yearly electric and thermal energy production, excluding the electric energy used for fluid pumping. The PV/T and separate PV and solar thermal collector plants were designed to produce the same amount of electric and thermal energy, so the denominator in the LCOE formula is the same. F_0 corresponds to the starting investment: the simplifying hypotheses of full investment payment and plant operation start in the same year ($t = 0$) are also assumed.

The initial investment was computed for PV and solar thermal collector plant size similar to the ones present at HE on the basis of real HE costs [28] and on market quotations. The investment for the PV/T solar plant was computed to give parity with both PV and solar thermal collector, given the same thermal and electric yearly production. The cost to be considered in the LCOE analysis is the operation and maintenance (O&M) cost. $C_0^{O\&M}$ is the O&M cost, which is evaluated mainly on the basis of the module cleaning operation cost observed during the HE plant monitoring and evaluated in €/m^2 of module gross area, and cleaning frequency [28]. An escalation rate $r_{O\&M}$ is considered to evaluate the increasing costs of O&M due to plant component (inverter, wiring, fluid pump, pipeline, etc …) wear and tear over the years. Table 4 summarises the main assumptions regarding economic assessment.

3. Results and discussion

3.1. PV/T solar plant electric performance analysis

The PV/T solar plant electric performance was monitored both with and without active cooling (i.e. with the primary circuit pump of Fig. 4, respectively, turned on or off) for two years from HE opening up to September 2015. The mean daily values for array yield Y_A and performance ratio PR of the PV/T solar plant are shown, respectively, in Figs. 7 and 8. Figs. 7 and 8 show that both Y_A and PR

Table 3
Flat plate solar thermal collector main characteristics: collector efficiency parameters are related to aperture area A_P.

Module characteristic	Symbol	Value
Dimensions (mm x mm x mm)	–	1700 × 1160 × 80
Gross area (m^2)	A_G	1.97
Aperture area (m^2)	A_P	1.89
Weight (kg)	–	35
Peak thermal efficiency (%)	η_0	0.768
Linear coefficient of thermal dispersion (W/m^2K)	a_1	3.570
Quadratic coefficient of thermal dispersion ($W/m^2 K^2$)	a_2	0.016
Incident Angle Modifier at $\Theta = 50°$ (%)	$k_\Theta(50)$	90
Cooling fluid flowrate (kg/s)	m_F	0.042
Pressure drop (mbar)	–	5

Table 4
Assumptions regarding economic assessment.

Item	Symbol	Value
PV plant investment cost (size under 5 kWp) [28]	F_0/kWp	2000 €/kWp
Solar thermal plant investment cost (size under 50 m^2)	F_0/m^2	750 €/m^2
Evaluation period	*n*	20 years
Discount rate	*i*	4.0%
Module cleaning cost (referred to A_G) [28]	–	1.5 €/m^2 (roof installation) 1.0 €/m^2 (ground installation)
Yearly inflation rate	*e*	1.0%
Yearly O&M cost escalation rate	$r_{O\&M}$	1.0%

of the PV/T solar plant are not much influenced by the active cooling functioning. In fact, the linear trends (continuous line with cooling, dotted line without cooling in both Figs. 7 and 8) are substantially the same, since the active cooling produces a benefit that is in the range 0–0.2 Wh/W for Y_A, and in the range 1–3% for PR, coherently with literature findings [16,35].

So, if the PV/T cooling system is set to produce domestic hot water, the expected benefits on PV electric performance due to module temperature T_{mod} decreasing are limited, because T_{mod} does not decrease appreciably. The difference between mean measured T_{mod} on PV/T solar modules with and without cooling is shown in Fig. 9. The data are taken from days with Y_R higher than 7, but at different mean ambient temperature T_a. Fig. 9 shows that module cooling reduces T_{mod} by about 1–2 °C, thus producing a mean 0.5–1.0% increase in PV electric power output. Therefore, if PV/T functioning is set for the production of domestic hot water, only a marginal benefit, in the order of a 2% increase in PV electric yield, can be achieved when active cooling is present. Otherwise, a higher increasing in the electric yield can be reached through the temperature decreasing of the cooling fluid (being the flowrate a constant), but in this case the water would be produced at too low temperature (i.e. under 40 °C) for any kind of use. In the case of active cooling mode functioning, the influence of mean cooling fluid inlet temperature T_{in} is also shown in Fig. 9: experimental results confirm Eq. (10) formulation, that is, at constant G, T_a variation has more influence than T_{in} variation for determination of T_m, which is clearly connected with the measured T_{mod} values.

Fig. 10 shows the mean daily Y_A and Y_R measured in the period from October 2014 to September 2015, expressed month by month. During the period taken into consideration not all the

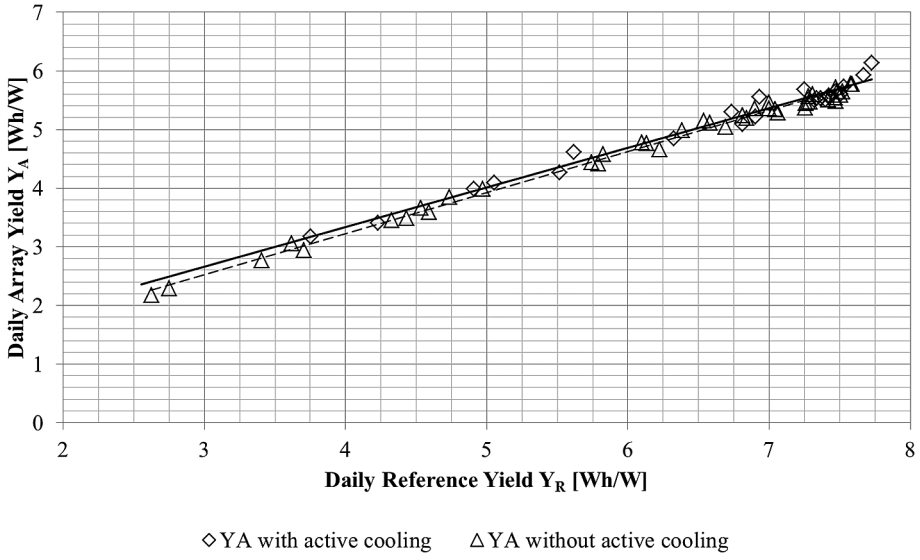


Fig. 7. PV/T daily array yield Y_A, with (continuous line) and without (dotted line) active cooling, plotted in relation with the daily reference yield Y_R.

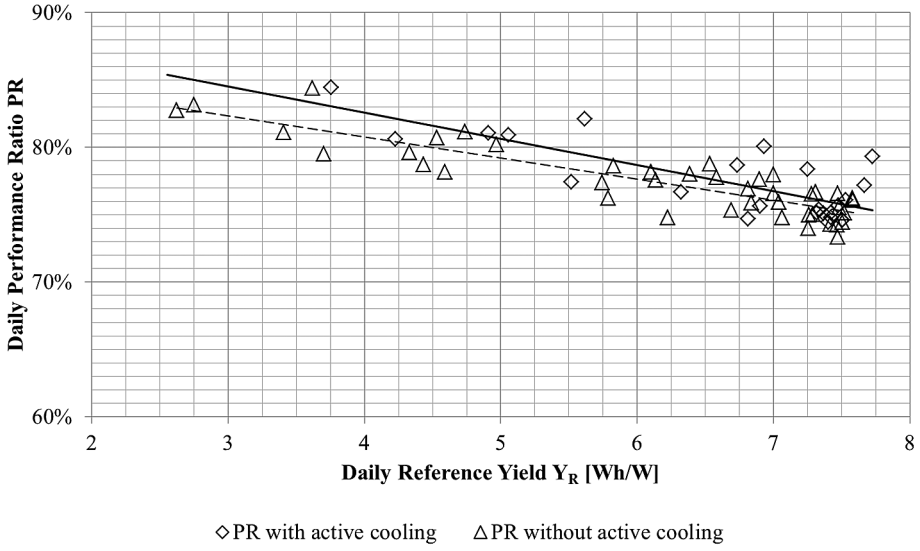


Fig. 8. PV/T daily performance ratio PR, with (continuous line) and without (dotted line) active cooling, plotted in relation with the daily reference yield Y_R.

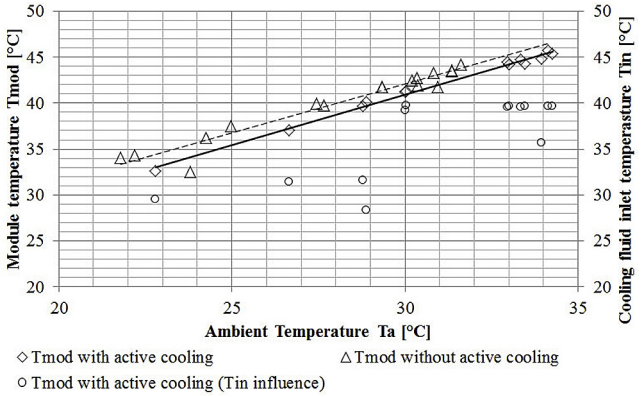


Fig. 9. PV/T daily mean module temperature T_{mod} as a function of mean ambient temperature T_a, measured daily reference yield Y_R higher than 7, with (continuous line) and without (dotted line) active cooling. For active cooling only, also the cooling fluid inlet temperature T_{in} is related to ambient temperature T_a (circular indicator).

meteorological data were available for the analysis [28]: to overcome this issue, meteorological data measured in the same period

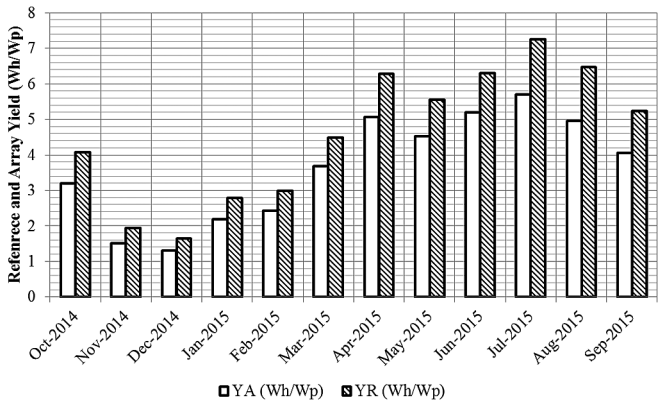


Fig. 10. Mean daily Y_R and Y_A from the PV/T solar plant without active cooling (only electric energy production) expressed month by month from October 2014 to September 2015.

were repeated to complete the database, thus allowing a yearly analysis.

The PV/T solar plant without active cooling (only electric energy production) produced 1335 Wh/Wp, with a mean annual PR of about 80%. On the basis of previous impact assessment of active cooling, a 2% electric energy yield can be foreseen for the same PV/T solar plant with active cooling functioning over the entire year: so, the PV/T solar plant with active cooling (both electric and thermal energy production) should produce 1362 Wh/Wp.

When the PV/T solar plant produces both electric energy and heat for domestic hot water purpose, no relevant increase in the electric performance of the PV modules can be observed. Therefore, the design of the electric subsystem of the PV/T solar system, i.e. the matching with the users and/or the design of electric batteries, can be substantially made by ignoring the effect of PV cooling on the electric yield of the PV/T solar system.

3.2. PV/T solar plant thermal performance analysis

The PV/T solar plant was tested in cooling mode from the end of May to the end of September 2015, when the most favourable conditions (i.e. high solar radiation and high ambient temperature) for heat recovery were present. The heat sink (dry cooler) activation temperature was set at 45 °C: so, until the storage tank reaches this temperature, the heat sink is turned off. During the test, the heat sink was always switched off, so no heat was removed from the storage tank (with the exception of heat losses). This testing configuration was chosen to simplify the analysis of PV/T performance and to allow the highest replicability of the data, since plant performances are not influenced by user behaviour.

PV/T solar plant thermal performance is more complex to analyse since thermal yield is influenced by many factors, as Eq. (10) shows. Fig. 11 shows solar global radiation G and ambient temperature T_a variation throughout the 5th June 2015, which is representative of a day with uninterrupted solar radiation. The measured thermal power P_h and the outlet temperature of cooling fluid T_{out} are represented. The primary circuit pump of the cooling fluid turns on early in the morning at 6:00, but the starting transient of the PV/T solar plant substantially ends only at 12:30, once the temperature difference between cooling fluid inlet and outlet is higher than 5 °C, as described before. Therefore, in the period 6:00–12:30 the PV/T solar plant produced less thermal energy than

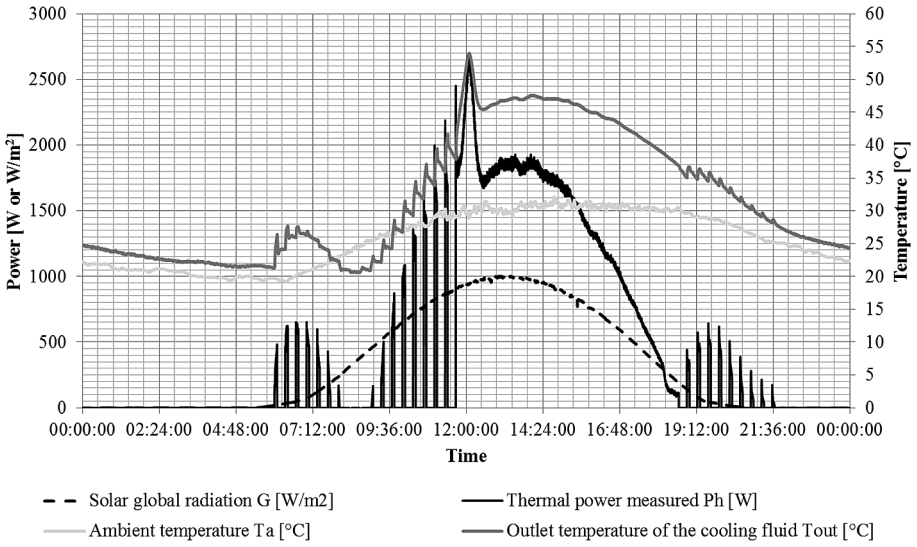


Fig. 11. Solar global radiation G, ambient temperature T_a, measured thermal power P_h and outlet temperature of the cooling fluid T_{out} during the 5th June 2015, which is representative of a day with uninterrupted solar radiation.

possible, since the primary pump was stopped every 15 min after 5 min working, waiting to reach the targeted temperature difference between cooling fluid inlet and outlet of 5 °C. So, the thermal yield of the PV/T solar plant is highly influenced also by the logic of the control system. In particular, heat recovery can be maximized through a reduction of the difference (down to 0.1 °C) between cooling fluid inlet and outlet which corresponds to the activation of the primary circuit pump; on the other hand, this kind of regulation implies the highest electric energy consumption for pumping. An equilibrium point should be identified, as it is dependent on plant characteristics, location and users. Starting from 12:30 and up to 18:30, the PV/T solar plant works in quasi steady-state condition.

Fig. 11 shows how both cooling fluid outlet temperature T_{out} and thermal power P_h follow the global solar radiation G trend: this is common behaviour which can be observed also on days with lower or more irregular solar radiation.

In the monitoring period mentioned before (May–September 2015), the PV/T plant produced a mean daily thermal energy of about 8.2 kWh th, with a maximum of 13.8 kWh th on 23rd July 2015 and a minimum of 1.7 kWh th on 17th June 2015. The first environmental parameter affecting thermal yield is the solar radiation G: Fig. 12 shows how daily thermal yield varies as a function of daily reference yield Y_R. An increase in solar radiation is usually accompanied by an increase in thermal yield, but as expected the relationship cannot be expressed by a linear trend, since a high degree of dispersion is present. The second environmental parameter to analyse is the ambient temperature T_a: once again, as expected, an increase in ambient temperature is usually accompanied by an increase in thermal yield, but with a high degree of dispersion with regard to a linear trend (Fig. 13).

Working condition impact on thermal yield was also evaluated. In particular, cooling fluid flowrate m_F varied from a mean daily value of 0.020 kg/s to 0.049 kg/s, which was below the maximum flowrate of 0.077 kg/s recommended by the PV/T solar panel supplier. The choice of the flowrate value has a potential impact on several factors. In fact, at constant environmental conditions (G and T_a) and cooling fluid inlet temperature T_{in}, a decrease in cooling fluid flowrate m_F produces an increase in mean temperature of the cooling fluid T_m, which means higher heat losses (see Eq. (8)). Moreover, a higher T_m can also produce a negative impact on the PV performance, since the temperature distribution across the PV modules is less uniform and can produce a decrease in the PV conversion efficiency. On the other hand, a lower m_F decreases electric energy consumption due to cooling fluid pumping. Conversely, an m_F increase reduces T_m, thus limiting heat losses as well as temperature non-uniformity across the PV panels, but

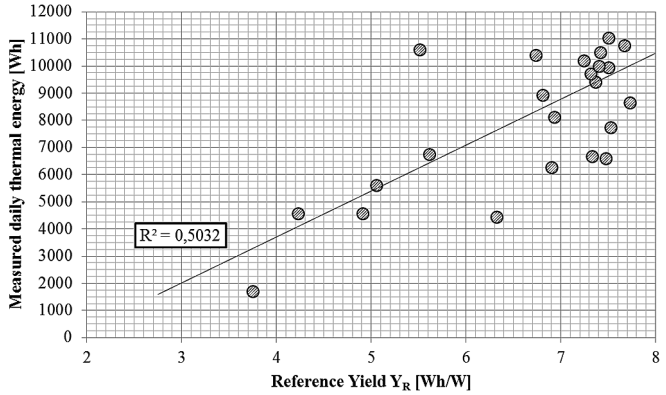


Fig. 12. Measured daily thermal energy produced by the PV/T solar plant as a function of reference yield Y_R.

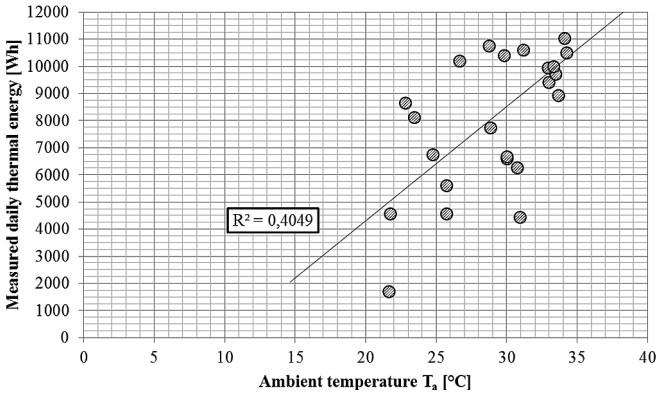


Fig. 13. Measured daily thermal energy produced by the PV/T solar plant as a function of daily average ambient temperature T_a.

increases pumping energy consumption.

Fig. 14 shows how daily thermal yield and array yield Y_A vary as a function of mean daily cooling fluid flowrate m_F; only days with Y_R higher than 7 were considered. No correlation can be found between thermal yield and cooling fluid flowrate variation, while array yield Y_A seems to be not significantly influenced by cooling fluid flowrate variation in the range taken into consideration.

The paper therefore shows that:

- seasonal thermal efficiency of the PV/T solar plant can be negatively affected by a non-optimized logic of the control system, which has to be opportunely designed, tested and implemented;
- thermal energy yield is strongly dependant on solar radiation and ambient temperature, but the relationship cannot be expressed through a simple equation;
- the cooling fluid flowrate has a relatively low relevance in the thermal performance of the PV/T solar plant.

3.3. ISO 9806 model validation

The thermal performance analysis also included the comparison between the measured thermal yield and the theoretical thermal yield. The theoretical thermal yield was computed through the application of Eq. (10), where environmental data collected by the local meteorological station (G, T_a) as well as cooling fluid flowrate m_F and inlet cooling fluid temperature T_{in} are considered as known parameters. PV/T solar thermal characteristics η_0, a_1, a_2 and A_A are indicated in Table 1, while k_Θ was considered equal to 1, thus ignoring incident angle variation influence on thermal efficiency.

Fig. 15 represents an example of how thermal power P_h was computed throughout the 5th June 2015, the same day taken into consideration in Fig. 11. In this case, measured daily thermal energy yield was 97% of theoretical daily thermal energy, computed according to Eq. (10). A good correspondence between measured and computed thermal energy was observed on days with uninterrupted solar radiation, wherein measured thermal yield was slightly lower than the theoretical one, probably due to incident angle variation influence that was not considered, in particular, in the morning and in the evening. So, an overestimation of solar thermal production can be expected through the application of the model described before.

When solar radiation G varies during the day, more relevant discrepancies can be found between measured and theoretical daily energy yield. In most of the cases (80% of monitored days), the measured thermal energy yield is lower than the theoretical one (with a minimum of 58%), but there are also days (20% of monitored days) wherein measured thermal energy yield was higher than the theoretical one (with a maximum of 130%). The cause of these variations is in the thermal inertia of the PV/T solar system, which was not included in Eq. (10).

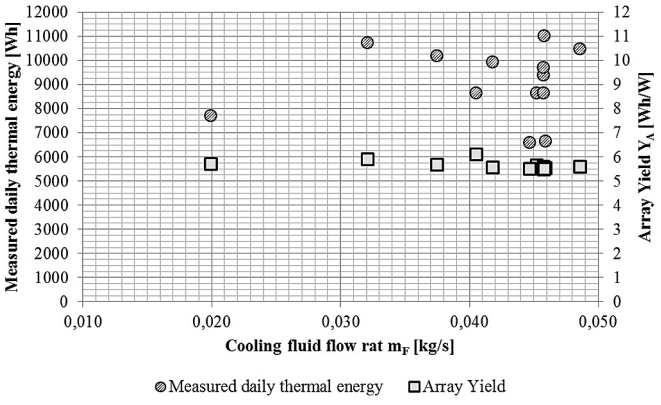


Fig. 14. Measured daily thermal energy and array yield Y_A produced by the PV/T solar plant as a function of cooling fluid flowrate m_F. Only days with reference yield Y_R higher than 7 were considered.

Thermal inertia is characterized by the thermal capacity C, measured in J/K. Thermal inertia contributes to the increase of heat losses thorough the term $-C \cdot dT_m/dt$: so, when T_m increases ($dT_m/dt > 0$), the thermal inertia contributes to reducing thermal power P_h. Vice versa, when T_m decreases ($dT_m/dt < 0$), the thermal inertia acts by increasing thermal power P_h. So, depending on how T_m varies although the day, depending on the environmental condition and/or operating condition, the measured thermal power P_h produced by the PV/T solar plant can be relatively lower or higher than the theoretical thermal power. On the other hand, if the whole monitoring period is considered, a theoretical mean daily thermal energy production of 9.1 kWh th can be computed, which is 11% higher than the measured one.

So, the ISO 9806 model (Eq. (10)) can be applied with good approximation to predict PV/T thermal yield in the medium-long term or for installations characterized by daily stable environmental conditions, since the presence of daily unstable conditions increases ISO 9806 model errors.

3.4. Electric and thermal yield comparison between PV/T solar plant and separate PV and solar thermal plants

The PV/T solar plant was compared with a separate PV plant (i.e. without active cooling) and a flat plate solar thermal collector plant (characteristics in Table 3) able to produce the same amount of thermal and electric energy, starting from the same environmental conditions measured by the local meteorological station. In particular, the configuration of the solar thermal collector was sized to annually produce the same thermal power at the same average temperature. The average temperature was computed as a weighted arithmetic mean by considering the daily thermal production as the weight as in Eq. (12).

$$T_{out_m} = \sum_{i=1}^{365} \frac{T_{out,i} \times E_{th,i}}{E_{th,i}} \tag{12}$$

To ensure that the two conditions were reached, a suitable configuration for the solar thermal collector plant has to be defined: to reach the same temperature as the PV/T solar system, a series connection is considered for the solar thermal collector modules, while to obtain the same thermal power without upsetting the mean outlet temperature, a parallel connection is considered.

In both systems (PV/T and solar thermal collector), it is presumed that the cooling pump starts only if the value of the solar radiation is higher than 300 W/m^2; moreover, if the outlet temperature is computed as lower than the inlet temperature of the cooling fluid, no thermal production is considered. During the period considered (October 2014–September 2015), not all the meteorological data were available for the analysis [28]: to overcome this issue, meteorological data measured in the same period were repeated to complete the database, thus allowing a yearly analysis, coherently with the electric yield comparison. The analysis was carried out by considering three different cooling fluid inlet temperature levels, 35 °C, 40 °C and 45 °C. Table 5 summarises the results of the solar thermal collector plants sizing process as previously described.

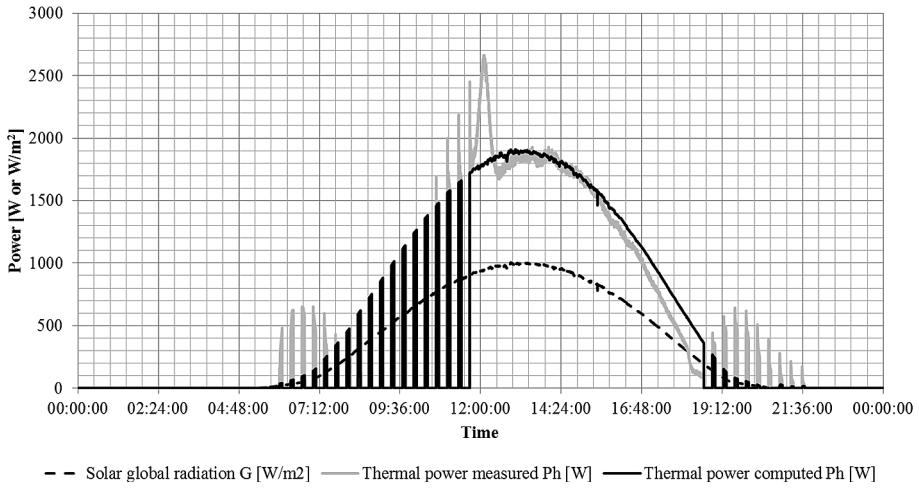


Fig. 15. Solar global radiation G, measured thermal power P_h and computed thermal power during the 5th June 2016, which is representative of a day with uninterrupted solar radiation.

Table 5
Solar thermal collector plants size as a function of inlet cooling fluid temperature and comparison with the PV/T solar plant: annual thermal yield, mean annual outlet temperature of cooling fluid, yearly water pump electric consumption.

Cooling fluid inlet temperature	T_{in} = 35 °C		T_{in} = 40 °C		T_{in} = 45 °C	
Hybrid system (PV/T) vs. Solar thermal collectors (TC)	PV/T	TC	PV/T	TC	PV/T	TC
Thermal energy production [kWh/year]	10,246	10,238	8081	8050	6176	6236
Water pump electric consumption [kWh/year]	9.3	0.3	8.4	0.3	7.3	0.2
Yearly mean cooling fluid outlet temperature [°C]	41.9	42.2	45.8	46.0	49.8	49.9
Number of solar thermal collectors in series	0	1.25	0	1.12	0	1.02
Number of solar thermal collectors in parallel	14	4.34	14	3.93	14	3.47
Total gross area A_G [m^2]	23.13	10.68	23.13	8.67	23.13	6.97

Table 5 underlines how, regardless of the cooling fluid inlet temperature, the solar thermal collector plant is able to produce the same yearly amount of thermal energy at the same temperature as the PV/T solar plant, occupying less area. In particular, the difference between the PV/T solar plant area and the traditional solar thermal collector plant area increases if the inlet temperature of cooling fluid increases. This trend is the consequence of the lower thermal dispersion of the flat plate solar thermal collector plant. Moreover, the PV/T solar plant electric energy consumption for cooling fluid pumping is considerably higher than the flat plate solar thermal collector plant consumption.

Fig. 16 shows how the thermal energy production and the mean cooling fluid outlet temperature vary by month. The trend is similar regardless of the inlet cooling fluid temperature: so, Fig. 16 shows the trends for the T_{in} = 45 °C, since it is the most interesting for domestic hot water production purpose. Fig. 16 shows how both plants concentrate energy production in summertime. On the other hand, the PV/T solar plant thermal yield difference between the period October–April and the period May–September is considerably increased. This is due to the higher thermal dispersion of the PV/T solar module, if compared with the traditional solar thermal collector. So, the flat plate solar thermal collector can produce a higher amount of domestic hot water than the PV/T solar system also in wintertime. Conversely, the PV/T solar plant has a higher production (at a higher temperature) in summertime. This behaviour should be considered when designing the PV/T solar plant, also taking the user characteristics into proper consideration.

Once the cooling fluid pump electric consumption had been computed, it was also possible to size the PV plant. The PV/T solar plant produces about 4377 kWh/year of electric energy, net electric

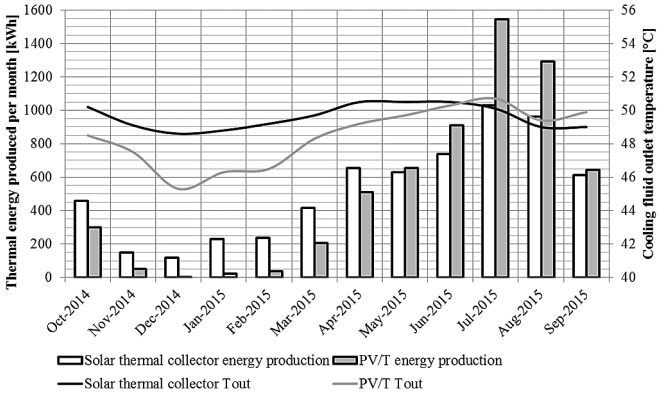


Fig. 16. Comparison between the solar thermal collector and the PV/T solar plant: thermal energy produced by month and mean cooling fluid outlet temperature, computed at a constant cooling fluid inlet temperature of 45 °C. Results correspond to systems of different areas but equivalent electrical and thermal outputs.

energy for cooling pumping. A PV plant of 3.28 kWp is needed to produce the same amount of electric energy. So, a preliminary relevant result is that the PV/T solar plant is able to produce the same amount of electric energy produced by the separate PV and flat plate solar thermal collector plant. In fact, PV/T and PV plant occupy more or less the same area, but PV and flat plate solar thermal collector plant occupy a larger area with regard to PV/T. The difference between PV/T area and PV plus solar thermal collector area decreases with inlet temperature T_{in} increasing, due to the thermal performance decreasing of the PV/T system (see Table 5).

So, the use of PV/T system, especially if working with users at relatively low temperature, produces relevant advantage when the area for plant installation is limited (i.e. roof installation).

The paper therefore shows that:

- the PV/T solar plant concentrates thermal energy production, both in terms of yield and higher temperature, in the period from May to September with a monthly heat peak production in July, which is 2200 times higher than the monthly heat production in December. The flat plate solar thermal heat ratio in the same months is only about 9. This great difference penalizes PV/T solar plant thermal effectiveness in comparison with flat plate solar thermal collectors if domestic hot water consumption is constant over the year (i.e. household), since it is more difficult to properly design the PV/T solar system;
- the PV/T solar plant is able to produce the same amount of yearly heat (at the same mean temperature) and electricity as the separate PV and flat plate solar thermal collectors, occupying less area.

3.5. Economics assessment

The thermal and electric yearly energy production of the PV/T and separate PV and solar thermal collector plants was estimated on the basis of data collected in the period from 1st October 2014 to 30th September 2015. On the basis of these data (see chapter 3.3) and of the economics assumptions summarised in Table 4, further considering the frequency of two complete cleaning operations of the solar modules per year, the LCOE (as defined in Eq. (11)) of the separate PV and solar thermal collector was computed at different cooling water inlet temperatures, resulting in 0.082 €/kWh (lower dotted line in Fig. 17) and 0.087 €/kWh and 0.092 €/kWh (upper dotted line in Fig. 17) at, respectively, 35 °C, 40 °C and 45 °C of cooling water inlet temperatures. Naturally, due to thermal energy production decrease, an increase in cooling fluid inlet temperature is accompanied by an increase in the LCOE. Fig. 17 shows the relationship between PV/T solar plant initial investment cost and LCOE, including also a differentiation on the basis of the cooling fluid inlet temperature considered.

A comparison was made between the PV/T solar plant and separate PV and flat plate solar thermal collector plants: parity was reached for an investment value in the PV/T solar plant that is near 3700 €/kWp, if the cooling water inlet temperature is considered to be equal to 45 °C (brown line in Fig. 17), or that is near 4700 €/kWp, if the cooling water inlet temperature is 35 °C (black line in Fig. 17). The economics analysis does not take into consideration the cost of the surface where the solar modules are installed: if a cost can be related to this issue too (i.e. land rent or purchase), the investment cost of a PV/T solar plant able to provide parity to separate PV and solar thermal collector plants will be higher since the PV/T solar plant occupies less area for the same thermal and electric produced energy.

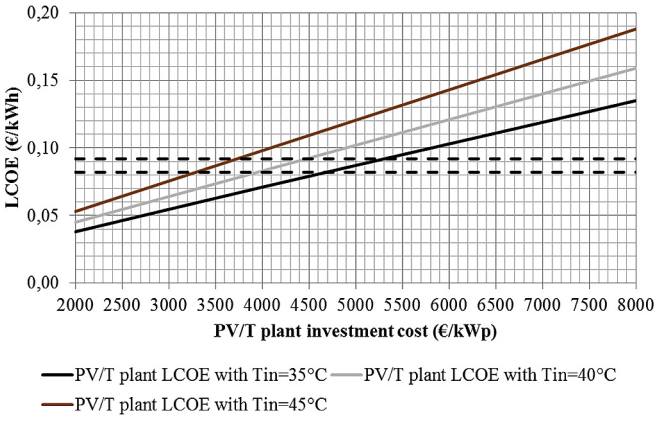


Fig. 17. Analysis of PV/T solar plant investment cost influence on LCOE (continuous line) and comparison with the parity level with highest and lowest LCOE of separate PV and solar thermal collector plants (upper dotted line for T_{in} = 45 °C, lower dotted line for T_{in} = 35 °C). The PV/T solar plant LCOE was differentiated on the basis of T_{in} value.

4. Conclusions

This paper shows the results of extensive monitoring activities carried out on a commercial PV/T solar plant at the HE outdoor development centre in Forlì (Italy) and reveals the potential offered by the PV/T technology in Central Italy (or, more in general, South-Central Europe). PV/T solar plant performances, both electric and thermal, were measured and compared in relation to different environmental conditions and to different working conditions, i.e. with or without cooling. Moreover, the ISO 9806:2013 standard model was applied to verify the estimation capacity of PV/T thermal performance, and the same model was then used to assess the yearly heat production of the PV/T solar plant. The notable points can be summarised as follows:

- the simultaneous production of electricity and domestic hot water through a PV/T solar plant does not have relevant effects on PV efficiency and yield, due to the relatively high temperature of the cooling water. So, for practical purposes (i.e. system design and matching with user characteristics) the electric behaviour of the PV/T solar plant over the year can be predicted with a good approximation, even without considering PV/T active cooling effects.
- The thermal performance of the PV/T solar plant is not greatly affected by cooling fluid flowrate.
- The temperature of hot water produced by the PV/T solar plant is compatible with domestic hot water production.
- Heat production at a relatively high temperature is concentrated in the months from May to September, while it is considerably lower in the rest of the year: domestic hot water production for users with relatively constant consumption over the years (i.e. households) does not seem to be a completely fitting application for PV/T technology, while it is more interesting for users with a concentrated consumption of domestic hot water in summertime (i.e. hotels, open air swimming pools, beach resort facilities).

A comparison between the PV/T solar plant and separate PV and flat plate solar thermal collectors was also carried out by considering both experimental (PV yield) and theoretical (thermal yield) performances. The comparison also includes the economics assessment through LCOE analysis. The PV/T solar plant achieved parity with an investment cost of about 3700–4700 €/kWp.

So, under certain economic conditions and after an in-depth analysis of user characteristics, PV/T technology can be a reliable solution for the combined production of electricity and thermal energy for domestic hot water production, with the great advantage, if compared with PV and solar thermal collector plants, of reduced area occupation, which can be a relevant selling point when the allowable surface for solar module installation is limited (i.e. roof installation).

On the other hand, it has been observed that a PV/T solar plant concentrates a large amount of the total yearly heat production in just a few months, while flat plate solar thermal collector heat production is more well distributed over the year, thus proving

more effective than PV/T technology when domestic hot water demand is constant (i.e. households). The next investigation will be focused on analysing the appropriateness of integrating a PV/T solar plant, instead of with a plant for the production of domestic hot water, with other technologies for the production of thermal energy for heating and/or cooling, in particular heat pumps and absorption chillers.

Acknowledgements

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